

Lab Experiment-2

```
[ ] pip install gensim

... Collecting gensim
  Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (8.4 kB)
Requirement already satisfied: numpy>=1.18.5 in /usr/local/lib/python3.12/dist-packages (from gensim) (2.0.2)
Requirement already satisfied: scipy>=1.7.0 in /usr/local/lib/python3.12/dist-packages (from gensim) (1.16.3)
Requirement already satisfied: smart_open>=1.8.1 in /usr/local/lib/python3.12/dist-packages (from gensim) (7.5.0)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart_open>=1.8.1->gensim) (2.0.1)
  Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (27.9 MB)
    27.9/27.9 MB 47.9 MB/s eta 0:00:00
Installing collected packages: gensim
Successfully installed gensim-4.4.0

[ ] import numpy as np
    from gensim.models import KeyedVectors

[ ] import gensim.downloader as api
    import numpy as np
    import matplotlib.pyplot as plt

[ ] model=api.load("word2vec-google-news-300")

[ ] [=====] 100.0% 1662.8/1662.8MB downloaded

[ ] x_gender=["man","male","boy","brother","he"]
    y_gender=["woman","female","girl","sister","she"]
    A_career=["career","business","salary","office","corporation"]

[ ] model=api.load("word2vec-google-news-300")

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[ ] x_gender=["man","male","boy","brother","he"]
    y_gender=["woman","female","girl","sister","she"]
    A_career=["career","business","salary","office","corporation"]
    B_family=["family","home","children","parents","marriage"]

[ ] x_european=["adam","grag","harry","josh","roger"]
    y_african=["jamal","leroy","tyrone","malik","darnell"]

[ ] def association(w, A, B, model):
    return np.mean([model.similarity(w, a) for a in A]) - \
           np.mean([model.similarity(w, b) for b in B])

[ ] def weat_score(X, Y, A, B, model):
    return sum(association(x, A, B, model) for x in X) - \
           sum(association(y, A, B, model) for y in Y)

[ ] def effect_size(X, Y, A, B, model):
    X_assoc = [association(x, A, B, model) for x in X]
    Y_assoc = [association(y, A, B, model) for y in Y]
```

```
[ ] def effect_size(X, Y, A, B, model):
    X_assoc = [association(x, A, B, model) for x in X]
    Y_assoc = [association(y, A, B, model) for y in Y]

    mean_diff = np.mean(X_assoc) - np.mean(Y_assoc)
    std_dev = np.std(X_assoc + Y_assoc)

    return mean_diff / std_dev
```

```
[ ] weat_gender = weat_score(x_gender, y_gender, A_career, B_family, model)
    effect_gender = effect_size(x_gender, y_gender, A_career, B_family, model)

    print("Gender WEAT Score:", weat_gender)
    print("Gender Effect Size:", effect_gender)
```

```
... Gender WEAT Score: 0.18831998
    Gender Effect Size: 0.34927812
```

```
[ ] if effect_gender > 0:
    print("Bias detected: Male terms more associated with Career")
elif effect_gender < 0:
    print("Bias detected: Female terms more associated with Career")
else:
    print("No detectable bias")
```

```
... Bias detected: Male terms more associated with Career
```

```
tests = {
    "Gender (Career vs Family)": effect_gender
}

for test, value in tests.items():
    print(f"{test}: {value}")
```

```
... Gender (Career vs Family): 0.34927812218666077
```

```
plt.bar(["Gender Bias"], [effect_gender])
plt.axhline(0)
plt.ylabel("Effect Size")
plt.title("WEAT Bias Comparison")
plt.show()
```

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```

...

